Quality Data Visualizations In Sports

A Literature Review

Cameron Scolari
Loyola Marymount University
Los Angeles, California, USA
cscolar2@lion.lmu.edu

Abstract

TODO

1 Introduction

Interaction design is a widely studied topic, but only recently have analysts used interaction design principles to guide their decisions when working with sports data. Visualizations allow us to investigate existing measures with targeted visualizations. A majority of data visuals are grounded in Tufte's foundational work, but most of them can be improved upon. Careful design of visualizations should tell the story of the data, highlighting important statistics and showcasing trends. With the growing number of visualizations being designed to represent the increasing availability of sports data, it is important that we understand what design techniques are standard across different sports, and where new designs can be developed. With this literature review, I analyze various designs across multiple sporting domains to provide a basis for which I should draw inspiration for my sports visualizations.

2 General Principles of Visual Design and Information Visualization

The Visual Display of Quantitative Information is a foundational work on data visualization and statistical graphics [16]. Tufte emphasizes how graphics should shift attention towards exploring the content of the data rather than towards questions of methodology. In other words, the graphs should explain themselves, highlighting the trends in data. Analysts should aim at giving viewers the greatest number of ideas in the shortest time with the least ink in the smallest space.

A central concept of Tufte's approach is the distinction between data-ink and non-data-ink. Data-ink is ink that carries valuable information, whereas non-data-ink consists of elements that do not contribute meaningful information to the graphic. He advocates for maximizing the amount of data-ink, and minimizing unnecessary elements. However, some words on a graphic can count as data-ink if they provide context and explain information.

Envisioning Information is a book that Edward Tufte wrote to introduce some general design rules [15]. He emphasized how color plays an important role in displaying information effectively. Color can be used to label, measure, represent or imitate reality, enliven or decorate, and make a section a focal point. However, color must be used carefully. He recommended not putting strong colors together as that can create unnecessary complexity. Strong colors are more effective when they are used individually to stand out from dull colors. Gray, for example, is a good background color as it allows for colors that represent measurements or labels to be prominent. However, color is not the only way to place emphasis on elements. Removing weight from graphs can help with distinguishing areas of greater importance from those with less importance.

Tufte also emphasizes that the visual complexity of a visualization should not exceed that of the dataset [16]. The number of dimensions used in a visual should not exceed the number of dimensions in the data. For example, using three dimensions to show only two variables adds complexity and can make interpretation more difficult.

Beyond the representation of data, Tufte discusses how the arrangement of graphical elements influence readability. When graphics function as lookup tables, grids can assist viewers in locating values and interpolating between them. Similarly, graphics can be reduced in size to compact more information within a limited space, provided that readability is maintained.

While visual design principles help determine how information is presented, the effectiveness of a visualization also depends on how users interact with it. In *Designing the User Interface*, Schneiderman identifies usability, usefulness, and universal usability as important qualities of effective interface design. He emphasizes the importance of understanding the user's needs and desired tasks before selecting an appropriate interaction style, such as direct manipulation, menu selection, form filling, or natural language. For interactive visuals, these are important because users should be able to explore information without needing to understand the complexity of the system and spend time learning the tools.

Direct manipulation is particularly useful in interactive visualizations because it allows users to interact directly with the data, allowing for filtering, highlighting, and other unique interactions. For example, Fu et al. incorporated sliders into their player profile visualizations to allow for users to dynamically filter information and adjust the data being displayed [3]. In another one of his works, Fu used direct manipulation over small multiples where highlighting one region in a single section highlighted the same region in all similar sections [4]. Vuillemot et al. used direct manipulation via drag and drop to allow users to drag teams into a bracket to make sports tournament predictions easier [17]. By allowing people to directly manipulate their predictions through dragging, users could predict the tournament in a non-linear fashion by predicting the "big picture" (tournament winner) first. Users could also easily change their predictions by simply dragging the team elsewhere in the bracket. Another example of direct manipulation is seen when Fu et al. allowed users to hover over NBA player faces to manually filter the shot maps to only show the shots taken by the hovered player [5].

Natural language interaction is something a bit more complex and not well studied in sports visualizations. This is because the data is so discrete and much better interacted with through filtering

using direct manipulation and menu selection. For example, typing which players you want to display graphics for is much more difficult than simply selecting them from a menu selection panel. However, menu selections can get complex fast. Schneiderman proposed that four to eight items per menu is the sweet spot. However, with hundred of players in the major sports leagues and dozens of teams, these menu selection panels can get quite large. This is where other interaction styles like form filling may be important. Instead of scrolling through a list of all MLB players, you could simply input the players name into a form box and quickly filter the menu.

Schneiderman also proposed eight golden rules of interface design to improve usability. These include maintaining consistency, supporting universal usability, providing informative feedback, designing interactions to prevent errors, allowing users to easily reverse actions, keeping users in control, and reducing the amount of information users must retain in short-term memory. These principles are particularly relevant to interactive data visualizations where users frequently adjust filters and explore different representations of data. For example, providing immediate feedback when a user changes a filter helps communicate how the visualization has been affected, while allowing actions to be reversed makes experimentation less restrictive. Similarly, maintaining consistency across controls and visual elements can help users understand how to interact with different parts of a visualization. Giving the users a sense of control makes them enjoy the graphic more as they do not have to struggle to interact with or understand the data they are visualizing.

In addition to providing design principles, Shneiderman identifies several measures for evaluating interface usability. These include the time required to learn an interface, speed of task performance, frequency of user errors, retention of knowledge over time, and subjective user satisfaction. These measures provide a framework for evaluating whether an interactive visualization is not only functional but also intuitive and effective for its intended users.

Similarly, Don Norman describes several principles of good interface design that emphasize making system functionality understandable to users [9]. These include making the current state of the system and available actions visible, providing users with a consistent conceptual model, establishing clear mappings between controls and their outcomes, and providing continuous feedback. These principles reinforce the importance of designing interfaces that communicate their functionality rather than requiring users to guess how interactions will affect the system.

Despite this, the beauty of design is that not all designers follow the same design principles and not all viewers enjoy the same designs. For example, Tufte profusely exclaimed how hatching in graphics looked bad explaining how specific areas of a graphic should be labeled with words rather than encoded with hatching. However, as seen in Figure 1, Perin et al. used hatching to visualize duel areas in soccer [11]. At the end of the day, it is all about making the data you have available work and designing for your users in the best way you see fit while trying to adhere to the general design principles that you follow.

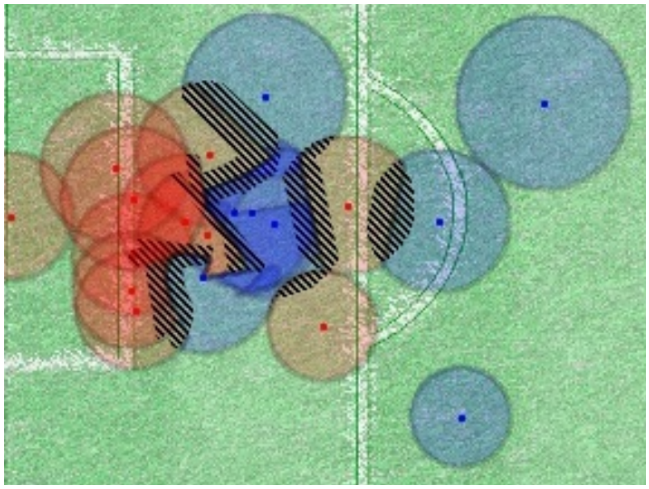


Figure 1: Hatching used to show duel areas in a soccer match.

3 Sports Data Visualization

Sports data visualization presents unique challenges due to the multidimensional and dynamic nature of sporting events. Sports data often involves spatial relationships, temporal sequences, player interactions, and performance measurements. Researchers have explored a variety of visualization techniques to help users analyze game events, understand player performance, identify trends, and compare teams and players. These approaches range from interactive visualizations that allow users to explore and manipulate data to traditional statistical graphics and spatial representations. This section examines existing approaches to sports data visualization, focusing on how different visual styles and interaction techniques have been used to communicate sports information.

One approach to communicating sports data is through automatically generated text, which can complement visual representations by providing relevant context and explanations. Perin et al. used automatic text generation to describe passes during a phase in soccer because analysts are used to text annotation [12]. Fu et al. used auto-generated text to describe statistics, providing another modality to interpret the data [4]. These approaches demonstrate how text can supplement graphics and provide alternative ways for users to understand sports data.

The dimensionality of graphics is another interesting topic because a majority of visualizations are just two dimensional. Designers can add more dimensions by varying the size and color of objects in a graph.

Dietrich et al. utilized color and shape to visualize the different locations of hard and easy catches in baseball games [2]. They followed multiple fundamental design principles by their choice of color. A majority of colorblind individuals mix up green and red, but by using blue and red to represent easy and hard catches, they allowed Figure 2 to be more accessible and usable to everyone.

Cox et al. were able to display tons of informative variables through color and a stacked bar chart [1]. As seen in Figure 3, the color indicates the outcome of a baseball game, the line on top encodes the absolute value of the run margin while the length

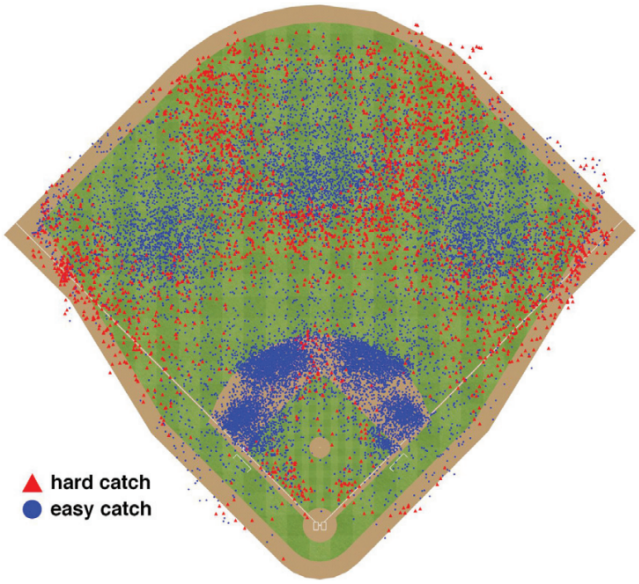


Figure 2: Graph showing the location of hard and easy players made in baseball.

below encodes the number of runs scored by the focus team. While they did not follow general usability principles and used green and red to encode wins and losses respectively, they did follow a different suggestion of Tufte. Tufte suggested removing black grids from the back of graphics because they add complexity [16]. He recommended to instead use white grid bars which this graph does because they still display the grid in a way that does not clutter the graphic.

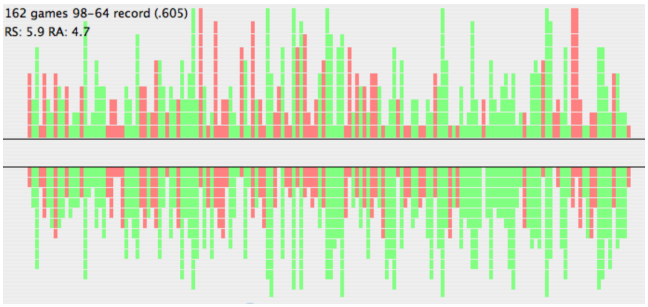


Figure 3: Graphic displaying the outcome and different metrics of games during a baseball season.

The dimensionality of a graphic can also be increased if you change your perspective of the data. Dietrich et al. developed Baseball4D, an interactive software, to support baseball game reconstruction and visualization [2]. The static nature of baseball allows for easy reconstruction of games. Using the data, they were able to create a three dimensional rendering of the game highlighting where players were positioned, where balls were hit, and more. The idea of reconstructing games from data to allow for later playback and review is fascinating, and something that can be adopted to other sports if play-by-play data is available. A similar approach is seen in Fu's work where they were able to showcase a three dimensional view of a basketball shot going into the basket [6]. This perspective is different from a traditional heatmap as it highlights how a shot may arc as it approaches the basket.

Perin et al. introduced a new visualization for visualizing the evolution of sports rankings over time called the gap chart as seen in Figure 4 [10]. One of their primary design goals was to ensure that tied entries never overlap to prevent loss of information. Unlike other ranking charts, the y-axis in their graph encodes both the rank and score with the score magnitude shown in gaps between lines. I see the benefits with this graph by grouping tied entries and separating entries by magnitude. However, I think as the number of lines increases, the graph gets cluttered and messy.

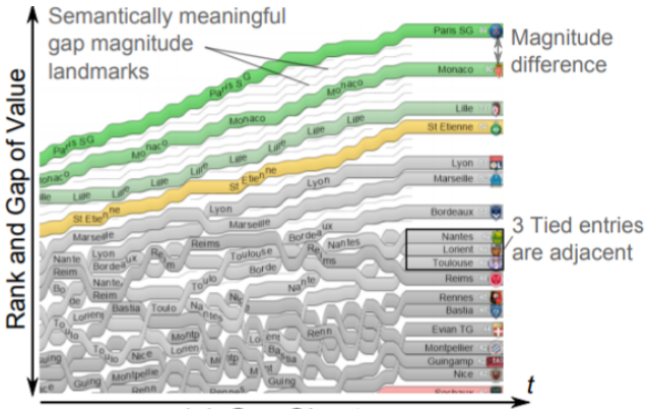


Figure 4: Gap chart showing soccer rankings over time.

Graphs do not have to be super complex to effectively display information. Some graphics providing dense info may not cater to casual fans and sometimes a simple table is the best way to communicate data. Though Cox voiced concerns about the cognitive effort required when reading lookup tables, tables are still utilized throughout sports to compactly display information [1]. Fu et al. used a table to display raw player stats [5]. As seen in Figure 5, common box score statistics are displayed in a table, with alternating rows being colored to allow for easier comparison among players. The aggregate totals are also displayed at the bottom.

Another common visual used across sports is the rectangle graph. Cox et al. used a rectangle plot to help better identify ineffective allocations of playing time to baseball players in Figure 6 [1]. The rectangle size is the number of plate appearances a player had during the season. The color can be changed to be any rate statistic like batting average. A player with a bigger rectangle but lower batting average may not be as effective as someone with a smaller rectangle but better average.

Another example of a rectangle graph is when Fu et al. used one for basketball to display the frequency of different shot types and the percentage the players make that shot with [5]. Figure 7 shows how color represents points per shot and size of rectangle encodes shot frequency. This graph is aggregated for a single team during a single game but it can be updated to represent a single player over a single game or over an entire season.

PLAYER	MIN	FG	FG%	3P	3P%	FT	FT%	eFG%	TS%	REB	AST	STL	BLK	TO	PF	PTS	+/-
Andrew Wiggins	40:22	7/16	43.8	1/6	16.7	3/4	75.0	46.9	50.7	7	2	2	2	3	4	18	-18
Draymond Green	34:47	1/4	25.0	0/2	0.0	0/0	0.0	25.0	25.0	4	3	0	1	2	6	2	-11
Kevon Looney	16:49	3/4	75.0	0/0	0.0	0/0	0.0	75.0	75.0	7	3	0	0	3	3	6	-8
Klay Thompson	39:02	7/17	41.2	5/13	38.5	6/6	100.0	55.9	63.6	3	3	1	1	0	1	25	-3
Stephen Curry	37:08	12/22	54.5	6/11	54.5	1/1	100.0	68.2	69.1	4	2	2	0	3	4	31	-11
Gary Payton	11:25	0/2	0.0	0/0	0.0	2/2	100.0	0.0	34.7	1	1	1	0	0	2	2	-14
Otto Porter	21:07	2/3	66.7	2/3	66.7	0/0	0.0	100.0	100.0	1	4	1	0	1	1	6	+3
Jordan Poole	24:28	4/8	50.0	1/4	25.0	1/2	50.0	56.3	56.3	3	3	0	0	3	2	10	-6
Andre Iguodala	02:09	0/0	0.0	0/0	0.0	0/0	0.0	0.0	0.0	0	0	0	0	0	0	0	-1
Nemanja Bjelica	03:27	0/1	0.0	0/0	0.0	0/0	0.0	0.0	0.0	0	1	0	0	0	0	0	-3
Moses Moody	02:19	0/0	0.0	0/0	0.0	0/0	0.0	0.0	0.0	0	0	0	1	0	0	0	-2
Jonathan Kuminga	02:19	0/0	0.0	0/0	0.0	0/0	0.0	0.0	0.0	1	0	0	0	0	0	0	-2
Damion Lee	02:19	0/1	0.0	0/1	0.0	0/0	0.0	0.0	0.0	0	0	0	0	0	0	0	-2
Juan Toscano-Anderson	02:19	0/0	0.0	0/0	0.0	0/0	0.0	0.0	0.0	0	0	0	0	1	0	0	-2
Total	48:00	36/78	46.2	15/40	37.5	13/15	86.7	55.8	59.1	31	22	7	5	16	23	100	-16

Figure 5: Table displaying common box score statistics for a basketball game.

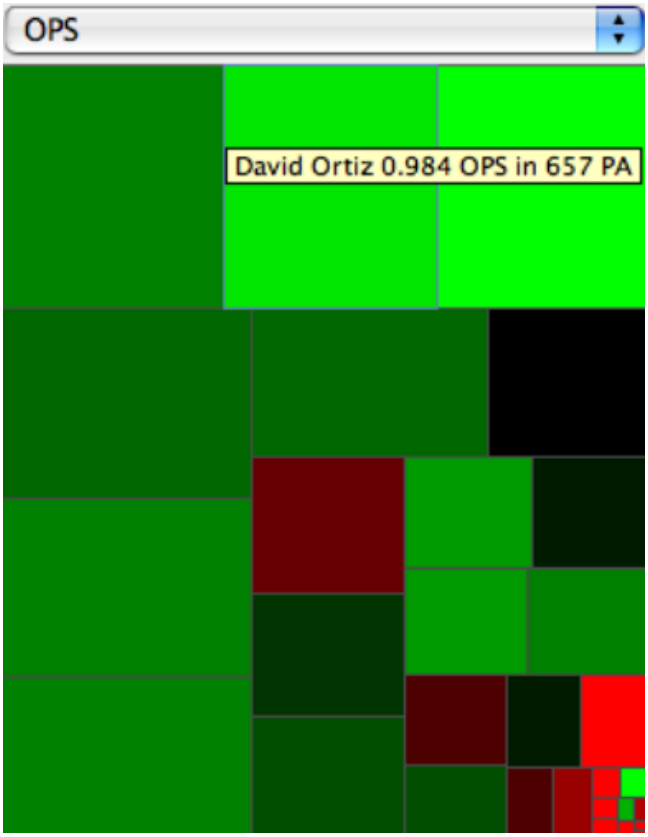


Figure 6: Player map for the 2004 Boston Red Sox colored by OPS.

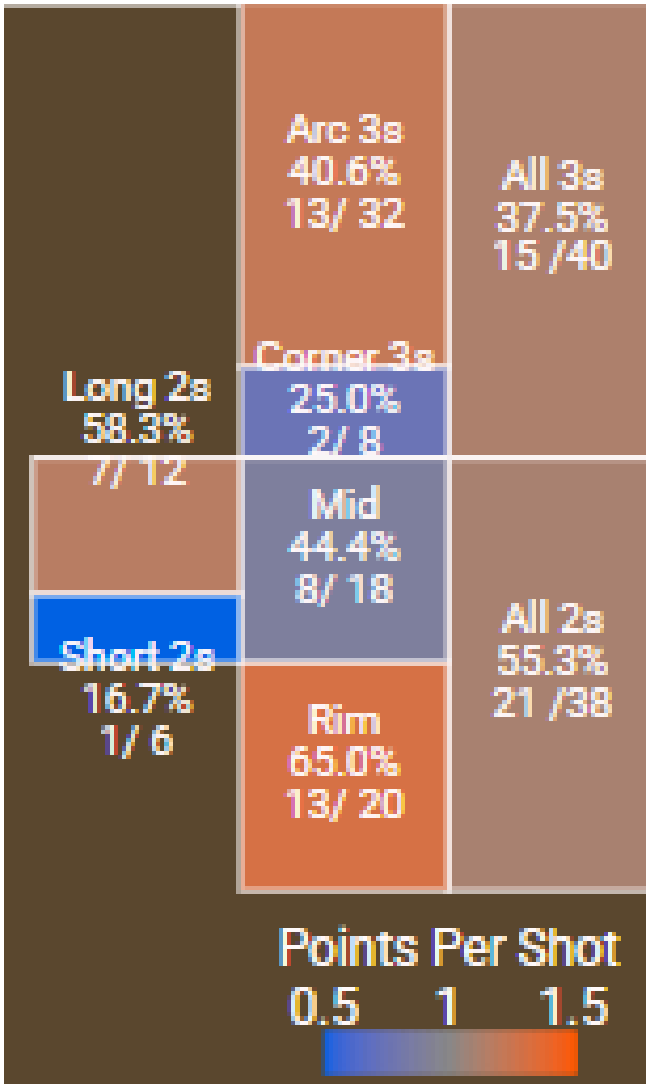


Figure 7: Rectangle plot showing the shot frequency and effectiveness for the Golden State Warriors against the Boston Celtics.

Another unique basketball graph is one using lineup rotation data to visualize the different stretches of time a player is on the court during a basketball game [5]. Figure 8 uses color to indicate the point differential during that time of the game, and the actual bars represent the time that player was on the court. This visual is good for scouting as it provides some information on when an opposing coach may make key substitutions during games.

Perin et al. used a heatmap with soccer player tracking data to visualize locations that players frequent often during a match [12]. Figure 9 shows how Messi spent most of his time in a single half of the field during this match, though whether he was home or away is ambiguous. Graphs like this can allow for opposing teams to develop more precise scouting reports and prepare better for upcoming games against him.

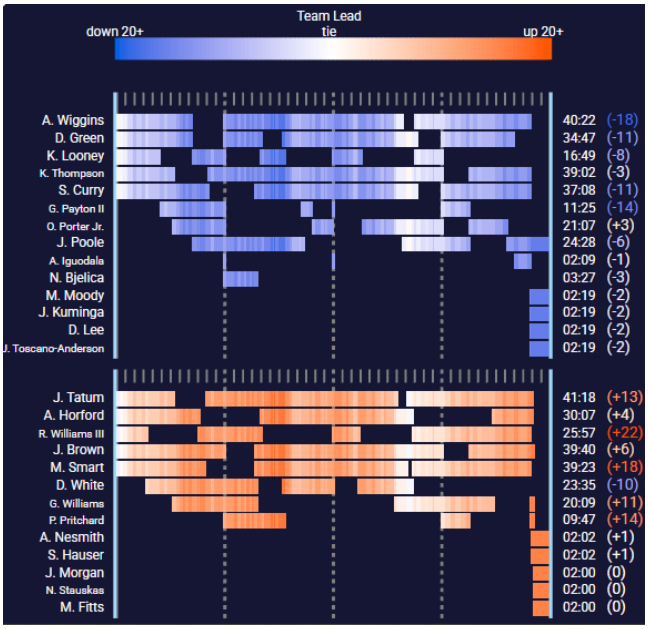


Figure 8: A team's player rotations during a basketball game.

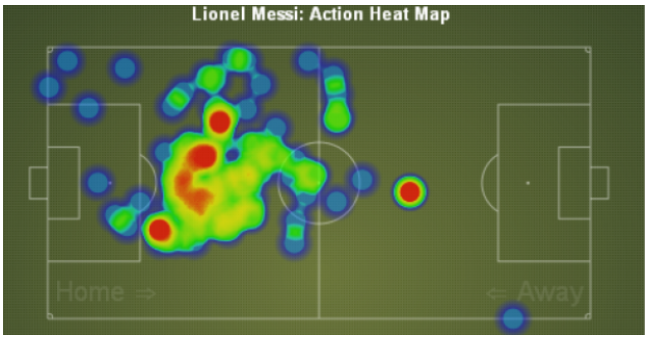


Figure 9: Heatmap of Messi's position during a game.

Common graphs like the scatterplot are also seen in sports visualizations. For baseball, analysts used a scatterplot to map the exit velocity and the launch angle and find a sweet spot for combinations that produce home runs as seen in Figure 10 [13].

Fu et al. used a scatterplot to highlight how different basketball lineups perform together [5]. This scatterplot, as seen in Figure 11, is a great way to analyze how effective lineups are together and compare them to the rest of the league. The size being the minutes the lineup has played together also helps suggest whether certain lineups should be played less or more.

A connected scatterplot can be used to show trends over time as displayed in Figure 12 [13]. This scatterplot shows how the average kick distance and field-goal percentage has increased over the years in the NFL.

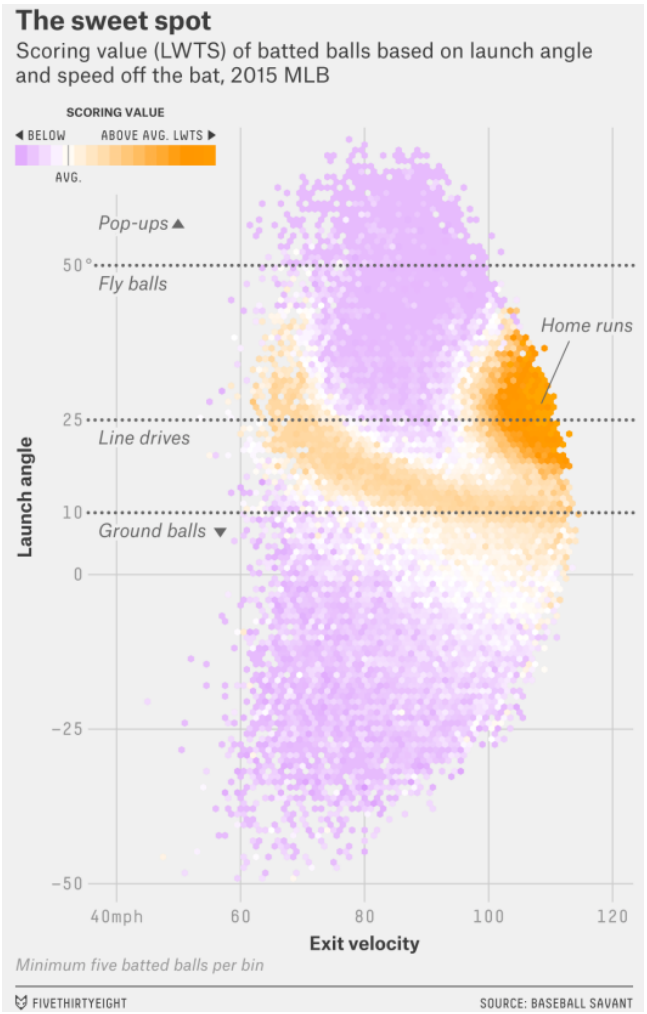


Figure 10: Scatterplot of exit velocity vs. launch angle to map the sweet spot for homeruns.

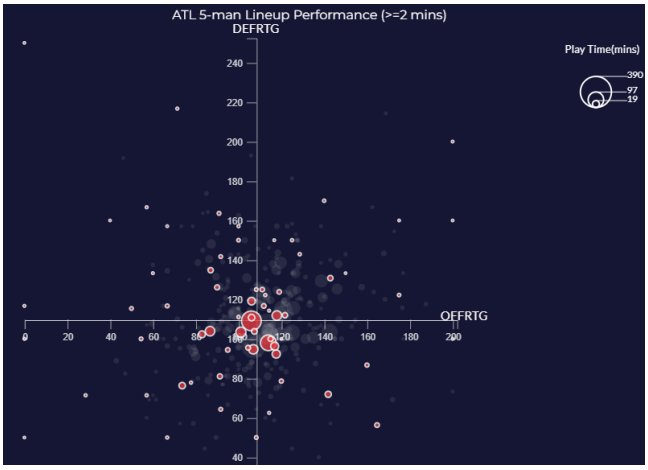


Figure 11: Atlanta Hawks 5-man lineups performance in offensive and defensive rating relative to other 5-man lineups across the league.

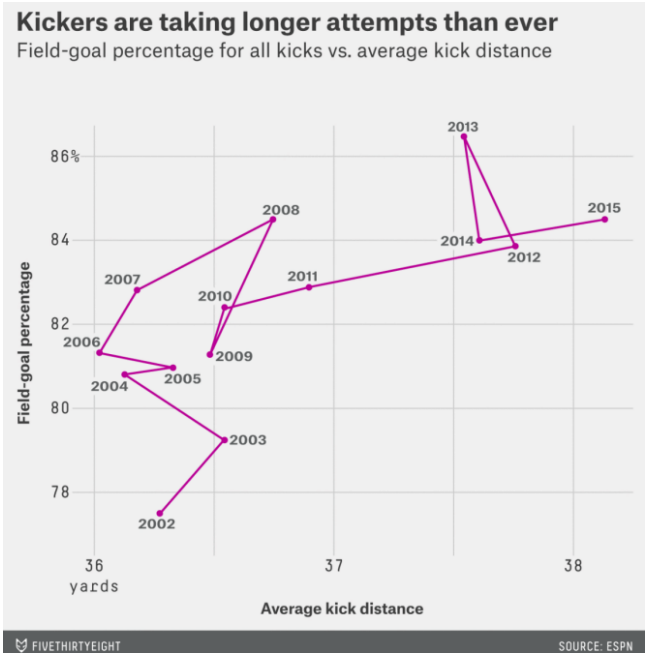


Figure 12: Connected scatterplot showing how kickers are taking longer attempts and making them more often.

4 Small Multiples for Sports Visualization

Small multiples, a visualization technique introduced by Edward Tufte, facilitate the comparison of multiple datasets through consistently structured visual representations [16]. Small multiples work as efficient and convincing summaries of data by enforcing direct visual comparisons across graphs sharing the same variables and visual structure, while varying one or more dimensions [15]. This direct comparison is something sports analysts appreciate as it allows for the effective assessment of player statistics instantaneously [14]. Small multiples can be seen in every major sport.

4.1 Examples Across Sports

As seen in Figure 13, Dietrich et al. used a heatmap and small multiples to highlight the way hits and fielder trajectories change depending on pitch type [2]. Aerial heatmaps like this one are not only seen in baseball.

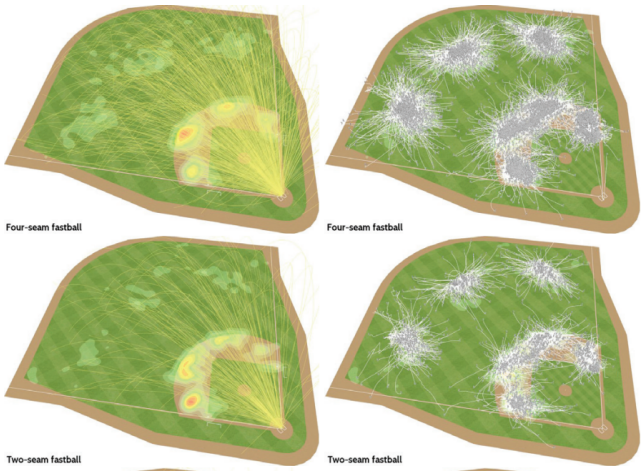


Figure 13: Ball and fielder trajectories vary for hits that arise from different pitch types.

Pileggi et al. used small multiples in hockey by visualizing a topdown heatmap view of a rink to map where shots occurred most frequently against specific defenses [14]. One of their design guidelines was that comparisons drive insights which is what led them to use small multiples. They worked with professional NHL analysts who expressed their appreciation for the small multiples.

Kirk Goldsberry used a similar shot heatmap small multiple to compare the shooting ranges and effectiveness of different basketball players [7]. As seen in Figure 14, size and color are used to highlight locations where players shoot the ball more ordinarily and their average points per attempt. This direct comparison helps analysts identify patterns and anticipate differences in shot-location distributions when facing different players. They can easily answer "where" questions like "where does Kobe Bryant score the most of his points from?"

Goldsberry also suggested the usefulness of overlaying heatmaps to analyze different matchups and observe unique trends. Fu et al. used side-by-side small multiples of basketball shot-chart heatmaps, with a central view highlighting the differences between the two [6]. This unique approach not only allows for the direct comparison between two players, but goes a step further by showing how they differ directly. Figure 15 shows how Fu used arrows, color, and size to indicate areas one player shot better or worse. This allows for easier understanding of more complex patterns like shot charts measuring a player's effectiveness when another player is on vs. off the court. It allows for direct comparison of seasonal trends and the comparison of players across different years. One pitfall with

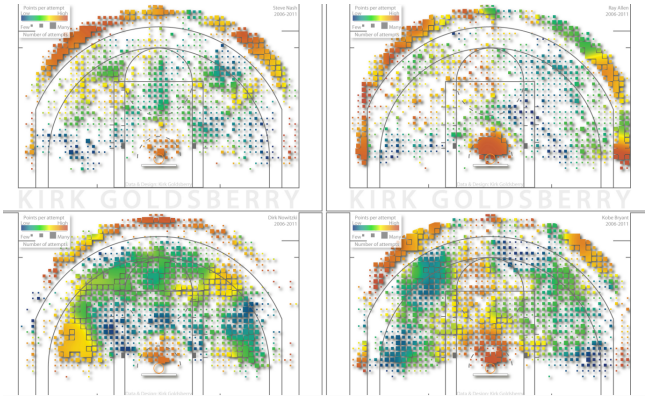


Figure 14: Shooting ranges of Steve Nash, Ray Allen, Dirk Nowitzki, and Kobe Bryant.

the central graph is the color palette for accessibility reasons we discussed earlier. Figure 16 and 17 shows how the visualization provides multiple view types all showing the same data to appeal to different user preferences, ensuring the user stays in control and allowing analysts of different levels to explore the data.

Fu et al. also worked on another basketball project that used small multiples [4]. Like many of the basketball small multiple examples, it was a shot heatmap to visualize the areas of the court where players shot the best. Each small multiple contained the players name and photo. In addition to the shot map, a pie chart was included to highlight the distribution of the locations players shoot from. One pitfall of the graph is that there is no clear legend to indicate what the colors represent, and some of the statistical abbreviations are not explained. A unique feature of the graph is that when one region is hovered in a single small multiple, it selects that region in all small multiples making direct comparison a lot easier. This is seen in Figure 18.

Figure 19 shows how Perin et al. used small multiples to map different phases of soccer plays [12]. This representation provides a good overview of the different phases or buildups a team used in a given match to attack on offense, allowing for a direct comparison among different attempts.

Kim et al. represented the draft history for each team in the NFL as small multiples utilizing color to indicate whether a draft pick is with a new team, out of the league, suspended, or with the team that drafted them [8]. The positions of each pick are displayed when a specific team's small multiple is selected, and hovering over a pick displays more about the player.

A unique example of small multiples is seen in Figure 21 where twitter data is displayed for each game in a tournament through a density plot with major events in the matches marked on the timeline [13]. These small multiples show the hype for different matches varies as the tournament progresses.

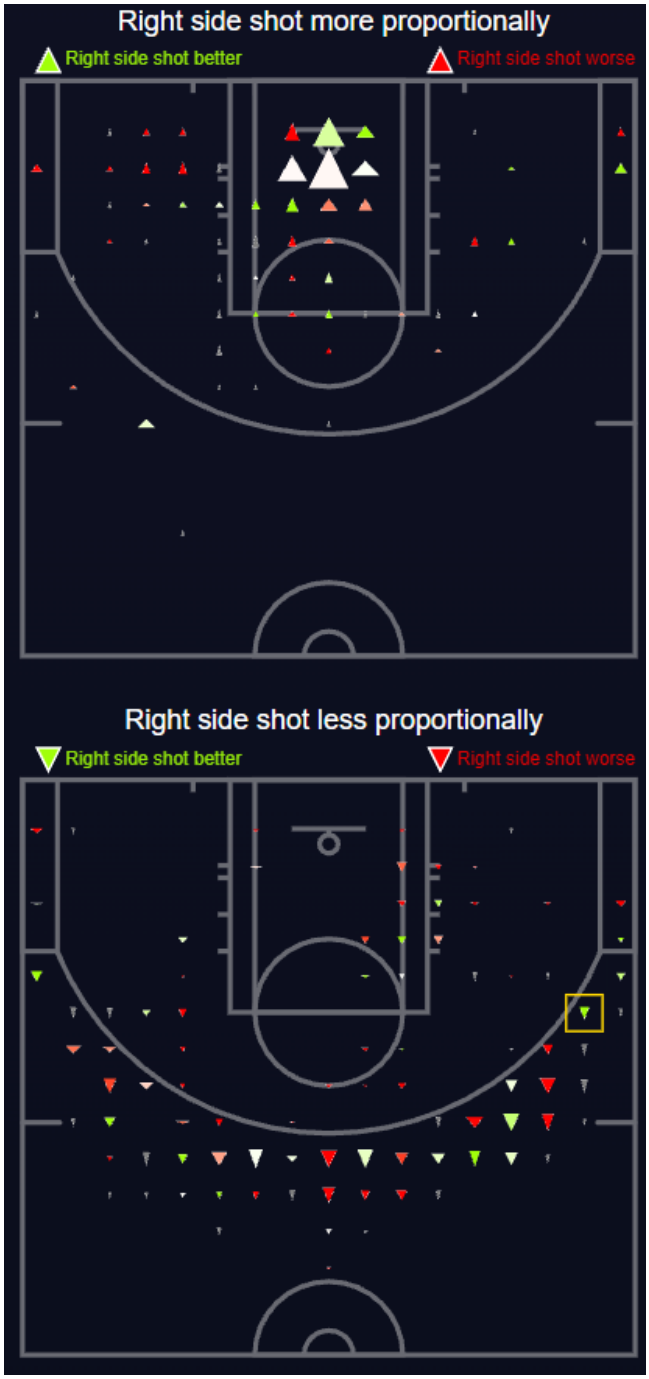


Figure 15: The central view showing the differences between two players.

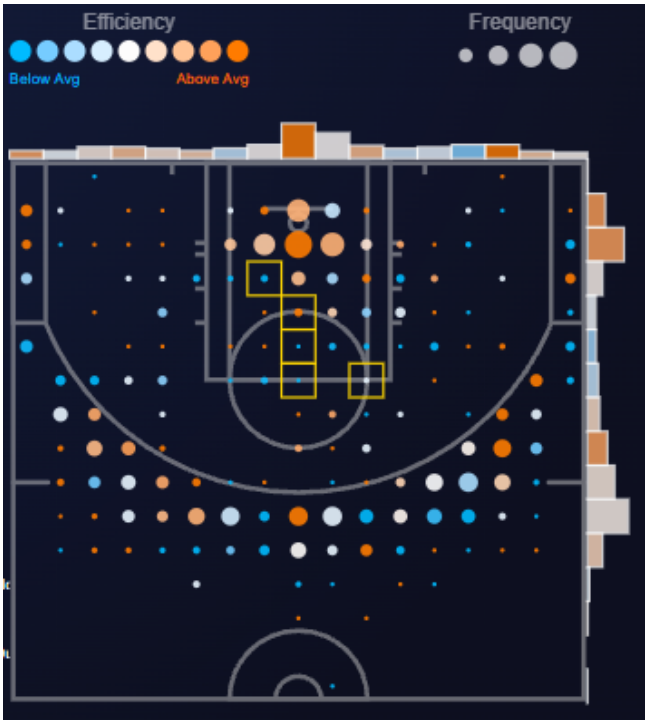


Figure 16: A players shot map with the cell view.

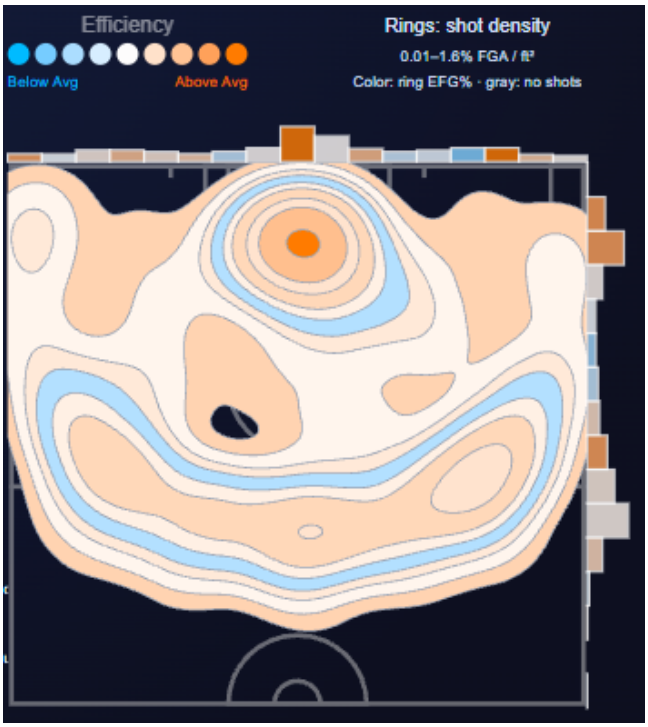


Figure 17: A players shot map with the contour view.

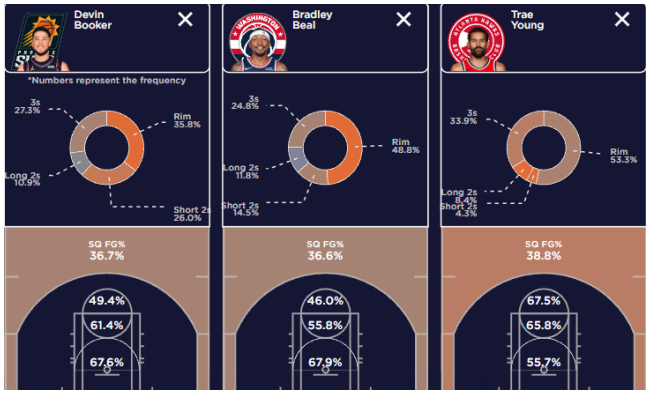


Figure 18: Player shot map small multiples with pie frequency chart.

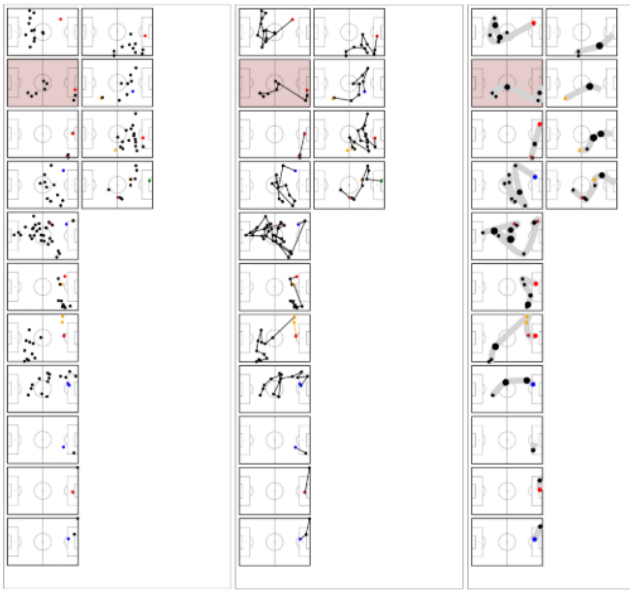


Figure 19: Small multiples of different soccer phases in a match.

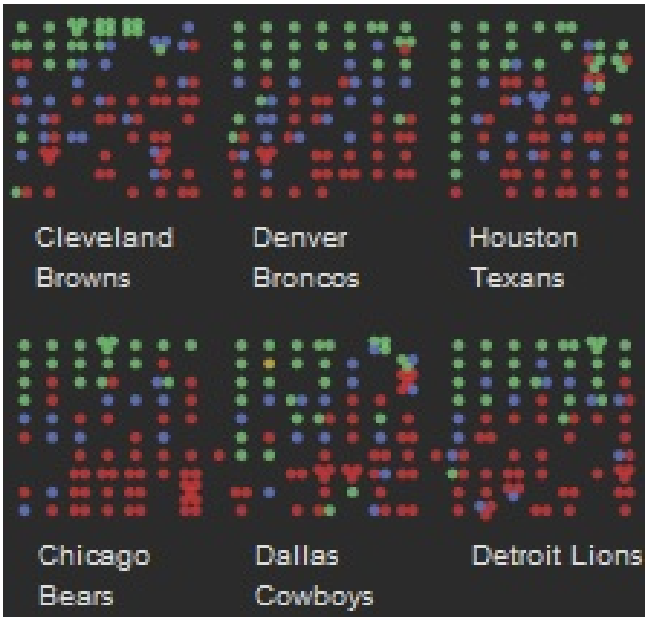


Figure 20: Small multiples for NFL teams and their draft picks.

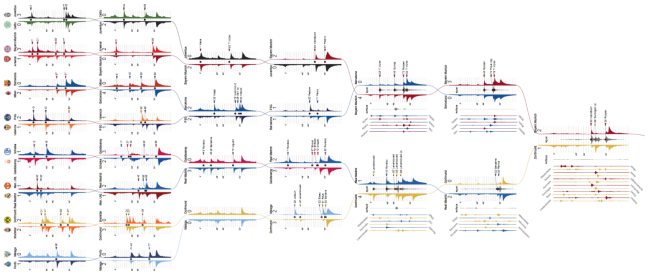


Figure 21: All the games of the UEFA Champion's League visualized using the tree structure of the tournament.

5 Discussion

TODO

6 Conclusion

TODO

References

[1] Andy Cox and John Stasko. 2006. SportVis: Discovering Meaning in Sports Statistics Through Information Visualization. In *Conference Compendium of the IEEE Symposium on Information Visualization*. 115–116.

[2] Carlos Dietrich, David Koop, Huy Vo, and Claudio Silva. 2014. Baseball4D: A Tool for Baseball Game Reconstruction and Visualization. *IEEE Transactions on Visualization and Computer Graphics* (2014).

[3] Yu Fu. [n. d.]. BballIndex Player Profile. Interactive basketball visualization. https://yufu-playerprofile.netlify.app/

[4] Yu Fu. [n. d.]. Shot Quality. Interactive sports visualization. https://shotquality-yufu.netlify.app/

[5] Yu Fu and John Stasko. 2022. Supporting Data-Driven Basketball Journalism through Interactive Visualization. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, Article 598, 17 pages. doi:10.1145/3491102.3502078

[6] Yu Fu and John Stasko. 2024. HoopInSight: Analyzing and Comparing Basketball Shooting Performance Through Visualization. *IEEE Transactions on Visualization and Computer Graphics* 30, 1 (2024), 858–868.

[7] Kirk Goldsberry. 2012. CourtVision: New Visual and Spatial Analytics for the NBA. In *MIT Sloan Sports Analytics Conference*. Boston, MA, USA.

[8] Se Yeon Kim, Sakshi Pratap, and John Stasko. [n. d.]. NFL Draft Vis: Exploring NFL Teams Draft History. Interactive sports visualization. https://sites.cc.gatech.edu/gvu/ii/sportvis/nfldraft/

[9] Donald A. Norman. 2013. *The Design of Everyday Things* (revised and expanded ed.). Basic Books, New York, NY.

[10] Charles Perin, Jonathan Boy, and François Vernier. 2016. Using Gap Charts to Visualize the Temporal Evolution of Ranks and Scores. *IEEE Computer Graphics and Applications* 36, 5 (2016), 38–49. doi:10.1109/MCG.2016.100

[11] Charles Perin, Romain Vuillemot, et al. 2016. Director's Cut: Analysis and Annotation of Soccer Matches. *IEEE Computer Graphics and Applications* (2016). doi:10.1109/MCG.2016.112

[12] Charles Perin, Romain Vuillemot, and Jean-Daniel Fekete. 2013. SoccerStories: A Kick-off for Visual Soccer Analysis. In *IEEE Transactions on Visualization and Computer Graphics*.

[13] Charles Perin, Romain Vuillemot, Charles Stolper, John Stasko, Jo Wood, and Sheelagh Carpendale. 2018. State of the Art of Sports Data Visualization. *Computer Graphics Forum* 37, 3 (2018), 1–24. doi:10.1111/cgf.13447

[14] Hannah Pileggi, Charles D. Stolper, John M. Boyle, and John T. Stasko. 2012. SnapShot: Visualization to Propel Ice Hockey Analytics. *IEEE Transactions on Visualization and Computer Graphics* 18, 12 (2012), 2819–2828. doi:10.1109/TVCG.2012.263

[15] Edward R. Tufte. 1990. *Envisioning Information*. Graphics Press, Cheshire, CT.

[16] Edward R. Tufte. 2001. *The Visual Display of Quantitative Information* (2 ed.). Graphics Press, Cheshire, CT.

[17] Romain Vuillemot and Charles Perin. 2016. Sports Tournament Predictions Using Direct Manipulation. *IEEE Computer Graphics and Applications* 36, 5 (2016), 62–71. doi:10.1109/MCG.2016.90